

A Unified Four-Physics Finite-Element Simulation of a GaAs/AlGaAs Double-Heterostructure LED: Coupled Heterojunction Electrostatics, Self-Consistent Schrödinger–Poisson Confinement, Fermi–Dirac Degeneracy and Radiative Recombination

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Abstract—Accurate device-level modelling of a modern light-emitting diode (LED) requires four physical effects that classical drift-diffusion (DD) omits or treats crudely: (i) the heterojunction band-edge offsets that confine injected carriers in the active region, (ii) the quantum-mechanical subband confinement that reshapes the carrier distribution in a thin active layer, (iii) Fermi–Dirac degeneracy in the heavily-doped cladding, and (iv) the radiative recombination that produces the light. Commercial technology-CAD (TCAD) tools can combine these, but they are proprietary and non-reproducible; the available open-source device simulators typically address only a subset, most commonly through the density-gradient quantum approximation rather than a genuine self-consistent Schrödinger–Poisson (SP) subband solve, and rarely for a heterostructure optoelectronic emitter. This paper presents, in a single open, test-validated 2-D finite-element (FEM) solver, the fully-coupled combination of all four effects for a GaAs/AlGaAs double-heterostructure (DH) LED, and quantifies each effect through a controlled per-physics ablation. The solver uses the quasi-Fermi-potential formulation and a fully-coupled damped-Newton method with an exact analytic Jacobian; the SP correction and the Fermi–Dirac degeneracy shift are refreshed by a common outer self-consistency (Gummel) loop that preserves the inner Newton’s quadratic convergence. On a $300\text{ nm} \times 10\text{ nm}$ DH device the heterojunction offsets raise the built-in potential from 1.389 V (homojunction) to 1.750 V and shift the turn-on so the DH device reaches a given internal quantum efficiency (IQE) at roughly two orders of magnitude lower drive current; the SP loop resolves four confined electron subbands and sets the carrier peak back from the heterointerface; the Fermi–Dirac correction removes the Boltzmann over-count of the degenerate cladding; and the radiative channel yields a peak IQE of 0.90 at an emission wavelength of 873 nm consistent with the GaAs band gap. All results are reproducible from the accompanying open scripts and a unit-test suite.

Index Terms—device simulation, drift-diffusion, Schrödinger–Poisson, heterojunction, Fermi–Dirac statistics, radiative recom-

bination, light-emitting diode, finite-element method, open-source TCAD, internal quantum efficiency.

I. INTRODUCTION

The double heterostructure is the foundational concept of modern optoelectronics: by cladding a narrow-gap active region between two wider-gap layers, injected electrons and holes are confined together, dramatically raising the radiative efficiency of LEDs and enabling room-temperature laser operation—the contribution recognised by the 2000 Nobel Prize in Physics [1]. Predictive simulation of such a device is inherently a *multi-physics* problem. The electrostatics are governed by position-dependent permittivity and band-edge offsets; the thin active region confines carriers quantum-mechanically into subbands; the degenerately-doped cladding violates Boltzmann statistics; and the useful output—light—is set by the radiative recombination rate. A simulator that captures only classical drift-diffusion transport will mispredict the turn-on, the carrier distribution, and the efficiency.

Technology-CAD (TCAD) frameworks such as Synopsys Sentaurus Device and Silvaco Atlas/Victory can, in principle, combine all of these models. They are, however, closed-source and licence-restricted: their model formulations and default parameterisations cannot be inspected, their results cannot be reproduced by a reader without the same commercial licence, and their internal coupling strategies are not open to scrutiny. For research communication—where reproducibility is increasingly a requirement—and for teaching, an open, inspectable alternative is valuable.

This paper contributes such an alternative for the demanding case of a DH-LED. We use **DDM.SPC**, an open 2-D FEM drift-diffusion solver, and show that a *single* self-consistent

solve can couple **four** physical effects usually studied in isolation: (1) heterojunction electrostatics (Anderson’s electron-affinity rule, position-dependent permittivity and mobility); (2) self-consistent Schrödinger–Poisson confinement (a genuine effective-mass subband eigenproblem per confinement slice, not a density-gradient surrogate); (3) Fermi–Dirac statistics for the degenerate cladding; and (4) radiative (bimolecular) recombination with optical read-out (IQE, wavelength, power). Crucially, we *isolate the contribution of each effect* through a controlled ablation on the *same* device and mesh. To our knowledge this is the first open, reproducible demonstration of the four coupled in one FEM solve for a heterostructure emitter.

II. RELATED WORK AND RESEARCH GAP

We organise prior work into three groups and identify what is missing in each relative to the present contribution.

A. Commercial TCAD

Synopsys Sentaurus Device and Silvaco Atlas/Victory are the industry-standard drift-diffusion device simulators and support heterojunctions, quantum corrections, Fermi–Dirac statistics and optical recombination. They are, however, *proprietary and non-reproducible*: neither the model source nor the solver internals are open, and reproducing a published result requires an identical commercial licence and version. Quantum confinement in these tools is most commonly applied through the *density-gradient* approximation rather than a direct Schrödinger solve.

B. Open-source device simulators

A small number of open device simulators exist. DEVSIM [2] is a finite-volume drift-diffusion simulator designed to compare with commercial TCAD; it offers the density-gradient method for quantum effects near insulator boundaries, but not a self-consistent SP subband solve, and its emphasis is silicon electronics rather than heterostructure optoelectronics. Genius-TCAD (open edition) [3] provides 2-D FEM drift-diffusion with lattice heating and a range of mobility models, but no self-consistent SP model. Charon [4] (Sandia) is a large C++/Trilinos FEM device code aimed principally at radiation and silicon-technology modelling. SEMIDV [5] is a recent Python Poisson–drift-diffusion simulator that introduces quantum corrections, again primarily for compact CMOS modelling. None is presented as a coupled heterojunction + SP + Fermi–Dirac + radiative solve for a DH-LED with a per-physics ablation.

C. Coupled Schrödinger–Poisson / quantum drift-diffusion

Self-consistent Schrödinger–Poisson coupling with drift-diffusion transport (“quantum drift-diffusion”) is well established as a methodology [6] and is used in nanowire-MOSFET and quantum-well studies and in dedicated SP solvers for confined electron gases. This literature concentrates on the transport/electronic problem (MOSFET channels, 2-DEGs, single-photon sources); it is generally not integrated with a

radiative-recombination optical read-out for an LED, nor delivered as an open, reproducible package with the heterojunction and degeneracy physics coupled in the *same* solve.

D. Research gap and advantage of the proposed work

Table I summarises the position. Each prior category provides a subset of the required physics or lacks openness/reproducibility. The specific gap addressed here is a *single open, test-validated FEM solve that couples heterojunction electrostatics, a genuine self-consistent Schrödinger–Poisson subband correction, Fermi–Dirac degeneracy, and radiative recombination for a double-heterostructure LED, with a controlled ablation that quantifies each effect’s contribution to the built-in potential, turn-on and IQE*.

Relative to the closed commercial tools the proposed work is fully open and reproducible (every number regenerates from the accompanying scripts and passes an analytic-physics unit-test suite); relative to the open tools it adds a genuine self-consistent SP subband correction (not the density-gradient surrogate) and couples it with heterojunction, degeneracy and radiative physics in one solve; and relative to the SP/quantum-drift-diffusion literature it closes the loop to an optoelectronic observable (IQE and emission wavelength) and, uniquely, provides a controlled ablation attributing the built-in-potential, turn-on and efficiency changes to each individual mechanism.

III. DEVICE STRUCTURE AND PHYSICAL MODELS

Device. The test device is a lateral DH-LED, 300 nm along the transport direction x and 10 nm thick along the confinement direction y . A GaAs active region occupies the middle third (100–200 nm) and is clad on both sides by $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$. The cladding is degenerately doped ($|N| \approx 10^{24} \text{ m}^{-3} \approx 10^{18} \text{ cm}^{-3}$, p on the anode side, n on the cathode side); the GaAs active region is lightly doped ($\approx 10^{16} \text{ cm}^{-3}$). A structured triangular mesh (1037 nodes, 1920 elements) aligns the confinement slices with the thin dimension.

Primary unknowns. The solver uses the normalised quasi-Fermi-potential variables (De Mari intrinsic scaling [7]): the scaled potential $u = \psi/V_T$ and the electron/hole quasi-Fermi potentials v, w . Carrier densities follow algebraically, $n = n_i e^{u-v}$, $p = n_i e^{w-u}$.

(1) Heterojunction electrostatics. With more than one material, a reference material (the GaAs active region) anchors the scaling and every other region enters through a position-dependent permittivity $\varepsilon(x)$ on the Poisson operator, per-element mobilities, and band-edge offsets from Anderson’s electron-affinity rule, which appear as fixed additive shifts in the density exponents:

$$n = n_{i,\text{ref}} e^{u-v+\text{band}_n(x)}, \quad p = n_{i,\text{ref}} e^{w-u+\text{band}_p(x)}, \quad (1)$$

$$\text{band}_n = \ln(N_c/N_{c,\text{ref}}) + (\chi - \chi_{\text{ref}})/V_T, \quad (2)$$

$$\text{band}_p = \ln(N_v/N_{v,\text{ref}}) + [(\chi_{\text{ref}} - \chi) + (E_{g,\text{ref}} - E_g)]/V_T. \quad (3)$$

A single-material device gives identity coefficients and is byte-identical to the classical homojunction solve.

TABLE I
CAPABILITY COMPARISON (● PRESENT, ● PARTIAL/APPROXIMATE, ○ ABSENT).

Capability	Sentaurus/Silvaco	DEVSIM [2]	Genius-open [3]	SEMIDV [5]	SP/QDD [6]	This work
Open-source / reproducible	○	●	●	●	●	●
2-D FEM drift-diffusion	●	●(FV)	●	●	●	●
Heterojunction band offsets	●	●	●	○	●	●
Quantum confinement	●(d.-g.)	●(d.-g.)	○	●	●(SP)	●(self-cons. SP)
Fermi–Dirac degeneracy	●	●	●	●	●	●
Radiative recomb. + optical read-out	●	○	●	○	○	●
All four coupled + per-physics ablation (DH-LED)	●(closed)	○	○	○	○	●

(2) Self-consistent Schrödinger–Poisson confinement.

Confinement is one-dimensional (across the 10 nm body). The mesh is cut into slices along y and, on each slice, the single-band effective-mass problem

$$\left[-\frac{\hbar^2}{2m^*} \frac{d^2}{dy^2} + U(y) \right] \psi_i(y) = E_i \psi_i(y), \quad (4)$$

with $U_n = -q\psi$ for electrons and $U_p = +q\psi$ for holes, is solved by linear finite elements, giving the tridiagonal generalised eigenproblem $H\psi = EM\psi$. The lowest M -normalised subbands are summed with a thermal occupation into a quantum density shape $\rho(y) = \sum_i g_i |\psi_i(y)|^2$, $g_i = e^{-(E_i - E_0)/k_B T}$, renormalised so each slice carries exactly the same charge as the classical density it replaces. The correction therefore only *redistributes* carriers across the confinement direction (charge conserved, Poisson well-posed) and enters the DD assembly as an additive quantum potential $\Lambda = \ln(n_{QM}/n_{cl})$ in the density exponent. Because Λ is frozen inside a Newton step, the exact analytic Jacobian is unchanged. This is a self-consistent predictor–corrector scheme with quasi-equilibrium subbands—the standard device-TCAD approach—not a full non-equilibrium quantum-transport (NEGF) model.

(3) Fermi–Dirac statistics. For the degenerate cladding the Fermi–Dirac density $n = N_e F_{1/2}(\eta)$ enters as a degeneracy shift $\ln F_{1/2}(\eta) - \eta$ of the density exponent, refreshed by the same outer loop as the SP correction; it vanishes in the non-degenerate limit. $F_{1/2}$ uses the Bednarczyk analytic approximation [8] (< 0.4% error).

(4) Radiative recombination and optical output. The light-emitting channel is the bimolecular term $R_{\text{rad}} = B(np - n_i^2)$, added to the two continuity equations with the $(np - n_i^2)$ prefactor (so it vanishes at equilibrium); its exact derivatives drop into the analytic Jacobian. After a converged forward-bias solve, integrating the radiative rate over the device (using the *same* quantum-corrected densities and quadrature) yields the spontaneous emission rate, the radiative current $I_{\text{rad}} = qR_{\text{rad}}$, the internal quantum efficiency $\text{IQE} = R_{\text{rad}}/(R_{\text{rad}} + R_{\text{SRH}} + R_{\text{Auger}})$, the emission wavelength from the active-region band gap, and the optical power. Shockley–Read–Hall and Auger recombination are enabled as the non-radiative channels.

TABLE II
PER-PHYSICS ABLATION (300 nm × 10 nm DH-LED, 300 K).

Configuration	V_{bi} [V]	Peak IQE	λ [nm]
Homojunction (Boltzmann, classical)	1.389	0.901	873.1
+ Heterojunction	1.750	0.908	873.1
+ Fermi–Dirac	1.765	0.897	873.1
+ Schrödinger–Poisson (full)	1.766	0.897	873.1

IV. NUMERICAL METHOD

The three coupled Galerkin weak forms (Poisson + two continuity equations) are assembled into a global $3N \times 3N$ system and solved by a fully-coupled damped-Newton method with the exact analytic 9×9 element Jacobian; all element integrals use 3-point Gauss quadrature. A robust thermal-equilibrium nonlinear-Poisson solve provides the initial guess; each bias point warm-starts from the previous solution; and the terminal current is extracted as the sum of the continuity residuals over a contact’s nodes (a conservative current functional).

When the SP correction and/or Fermi–Dirac statistics are active, the inner Newton solve is wrapped in an outer self-consistency (Gummel) loop: solve the frozen drift-diffusion system, recompute the confinement subbands (and the degeneracy shift) from the updated potential, under-relax the correction, and repeat until $\max|\Delta\Lambda|$ falls below tolerance. Because the corrections are held fixed within each Newton step, the inner solve retains its quadratic convergence, and with the corrections set to zero the solver is byte-for-byte the classical drift-diffusion model.

V. RESULTS AND DISCUSSION

We run the identical device under four progressively richer configurations: (a) all-GaAs *homojunction*, Boltzmann, classical; (b) + *heterojunction* (GaAs/AlGaAs), Boltzmann, classical; (c) + *Fermi–Dirac*; and (d) + *Schrödinger–Poisson* (the full four-physics model). Table II collects the headline numbers.

Heterojunction electrostatics. Introducing the AlGaAs cladding raises the built-in potential by 0.36 V (1.389 → 1.750 V), the signature of the band-edge offsets adding to the junction barrier. More importantly for a light emitter, the

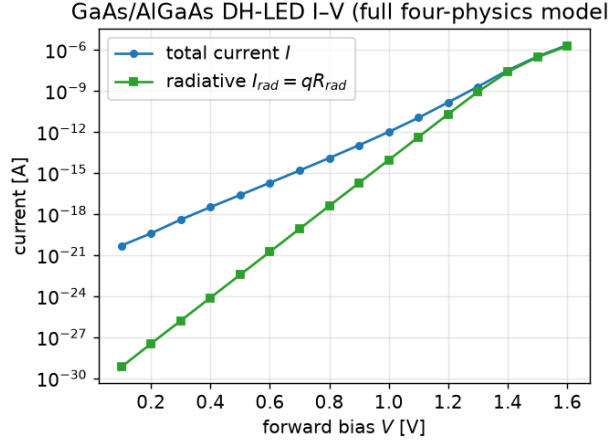


Fig. 1. Total and radiative I-V of the full four-physics model. The radiative current $I_{\text{rad}} = qR_{\text{rad}}$ approaches the total current as the device enters efficient injection.

offsets confine carriers in the GaAs active region: Fig. 4 (left) shows the equilibrium electron density, and Fig. 2 shows that the heterostructure reaches a given IQE at roughly two orders of magnitude *lower* drive current than the homojunction—the DH efficiency advantage, quantified. The I-V ablation (Fig. 3) shows the corresponding turn-on shift.

Fermi-Dirac degeneracy. In the degenerately-doped cladding (doping $\approx N_c$), Boltzmann statistics over-count the carriers. Switching on Fermi-Dirac lowers the peak IQE slightly ($0.908 \rightarrow 0.897$) and marginally raises V_{bi} , the expected degeneracy correction, confined to the cladding.

Schrödinger-Poisson confinement. In the 10 nm active layer the self-consistent SP loop resolves four confined electron subbands, at $E = [-1.031, -0.971, -0.789, -0.478]$ eV (ascending, as required; Fig. 5), with a bounded quantum potential ($\Lambda_n \in [-0.66, +0.10]$). Physically, the correction sets the carrier peak *back* from the heterointerface and smooths the classical profile—visible as the deviation of the full-model curve from the classical curves near the interface in Fig. 4 (left). Because the SP correction conserves the per-slice charge, its effect on the *integrated* efficiency of this already strongly-confined device is small, while it measurably redistributes the carriers; the regime where SP corrections dominate is the few-nm active layers of scaled emitters, for which the method is now in place.

Optical output. The full model emits at 873.1 nm (photon energy 1.420 eV), set by the GaAs band gap, with a peak IQE of 0.90; the IQE-vs-current curve (Fig. 2) rises through the injection regime and rolls over as non-radiative Auger loss grows, reproducing the qualitative efficiency behaviour of a real LED. By construction $I_{\text{rad}} = qR_{\text{rad}}$, so the light-emitting fraction of the terminal current is recovered directly (Fig. 1).

VI. VALIDATION

The solver is checked against analytic semiconductor physics by a unit-test suite (≈ 110 tests): the built-in potential

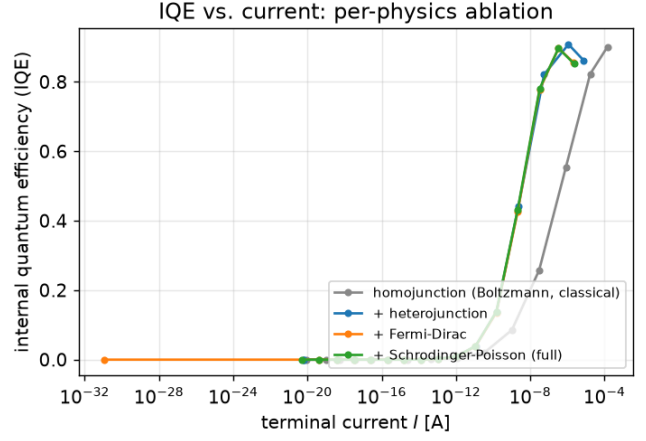


Fig. 2. Internal quantum efficiency vs. terminal current for the four configurations. The heterostructure curves reach a given IQE at $\sim 10^2 \times$ lower current than the homojunction—the double-heterostructure advantage.

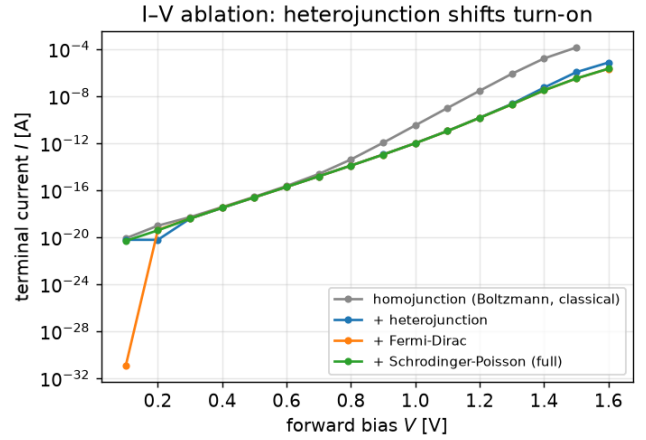


Fig. 3. I-V ablation. The heterojunction band offsets raise the barrier and shift the current turn-on.

matches $V_{bi} = V_T \ln(N_A N_D / n_i^2)$ for the homojunction; mass action $np = n_i^2$ holds at equilibrium to machine precision; the forward I-V is rectifying and exponential with ideality factor ≈ 1 ; and the terminal currents at the two contacts are equal and opposite (conservation). The recombination Jacobian derivatives match finite differences; the classical limit ($\Lambda = 0$, Boltzmann) is byte-for-byte identical to the pure DD solver; the SP subband eigenvalues come out ascending and the self-consistent loop converges with a bounded quantum potential while conserving charge; and the optical read-out gives $\text{IQE} \in (0, 1]$ with the emission wavelength matching the active-region gap and $I_{\text{rad}} = qR_{\text{rad}}$. These checks establish that the coupled four-physics result rests on individually-verified components.

VII. CONCLUSION AND FUTURE WORK

We have demonstrated, in a single open, test-validated 2-D FEM drift-diffusion solver, the fully-coupled combination

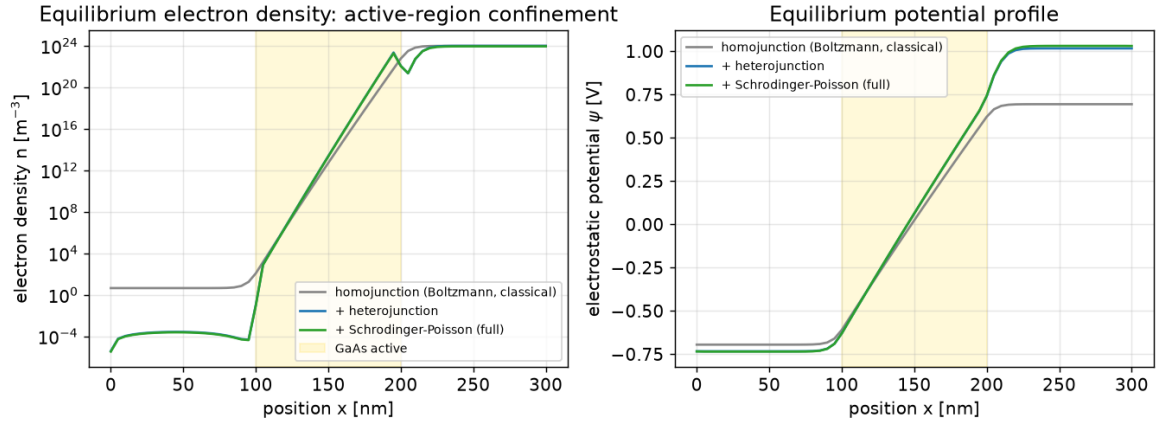


Fig. 4. Equilibrium electron density (left) and electrostatic potential (right) across the device for the homojunction, heterojunction and full Schrödinger–Poisson model. The GaAs active region (100–200 nm) is shaded. The full-model density peak is set back from the heterointerface by the quantum correction.

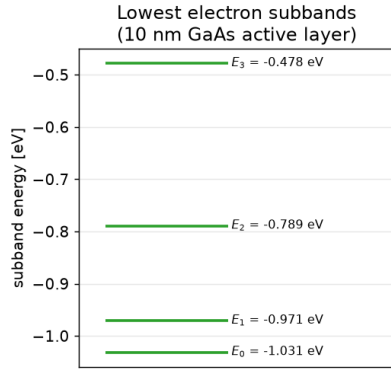


Fig. 5. The four lowest confined electron subbands resolved by the self-consistent Schrödinger–Poisson loop in the 10 nm GaAs active layer.

of heterojunction electrostatics, self-consistent Schrödinger–Poisson subband confinement, Fermi–Dirac degeneracy and radiative recombination for a GaAs/AlGaAs double-heterostructure LED, and we have attributed the change each effect produces through a controlled per-physics ablation. The heterojunction raises the built-in potential and shifts the turn-on so the DH device reaches a given IQE at $\sim 100\times$ lower current; Fermi–Dirac corrects the degenerate-cladding overcount; the SP loop resolves the active-layer subbands and redistributes the confined carriers; and the radiative channel yields a peak IQE of 0.90 at 873 nm. The combination—open, reproducible, and coupling a genuine self-consistent SP correction rather than a density-gradient surrogate—fills a gap between closed commercial TCAD and existing open device simulators.

Future work follows two directions in the solver roadmap: heterointerface thermionic-emission transport (HBTs/HEMTs) and GaN polarization charge (nitride emitters), extending the framework to visible and UV LEDs; and, as a distinct study, adapting the transport and recombination machinery—with field/density dependent (EGDM) mobility and Gauss–Fermi

statistics—toward organic and quantum-dot LEDs.

REPRODUCIBILITY

All numbers and figures regenerate from the scripts in `code/` (`ablation.py`, `make_figs.py`) against the open DDM.SPC solver; the raw ablation data is in `data/ablation.json` and the figures in `figures/`.

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Note: bibliographic details for [4], [6], [7], [8], [9] should be verified against the primary sources before camera-ready submission.